

EDINBURGH COURSEWORK / REVISED EXPOSITION

Flexes of a Smooth Cubic

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The Hessian criterion and a Hesse-family calculation

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1 The Hessian criterion for a flex

Work over the complex numbers. Let $C = \{F = 0\} \subset \mathbb{P}^2(\mathbb{C})$ be a smooth, irreducible plane cubic, where F is homogeneous of degree three. At $P = [a : b : c] \in C$, its tangent is

$$F_x(P)x + F_y(P)y + F_z(P)z = 0.$$

A *flex* is a point where the tangent has intersection multiplicity three with C . Multiplicity is essential: merely writing that the tangent “meets the curve at P ” does not distinguish a flex from an ordinary point. Define the Hessian polynomial

$$H_F = \det \left(\frac{\partial^2 F}{\partial x_i \partial x_j} \right)_{i,j=1}^3.$$

Proposition 1.1 (Local Hessian criterion). *A point P on a smooth irreducible plane cubic is a flex if and only if $H_F(P) = 0$.*

Proof. An invertible linear change of homogeneous coordinates puts P at $[0 : 0 : 1]$ and its tangent at $y = 0$. This does not change whether the Hessian vanishes: if $G(u) = F(Au)$, then

$$\text{Hess } G(u) = A^\top \text{Hess } F(Au)A, \quad H_G(u) = (\det A)^2 H_F(Au).$$

In the new coordinates set $a = F_y(P) \neq 0$. We have $F_x(P) = F_z(P) = 0$. The complex implicit function theorem gives a holomorphic function $y = \phi(x)$ near 0 with

$$F(x, \phi(x), 1) = 0, \quad \phi(0) = \phi'(0) = 0.$$

Differentiating this identity twice gives

$$F_{xx} + 2F_{xy}\phi' + F_{yy}(\phi')^2 + F_y\phi'' = 0.$$

At P this reduces to $F_{xx}(P) = -a\phi''(0)$. Differentiating Euler’s homogeneous identity $xF_x + yF_y + zF_z = 3F$ yields

$$xF_{xi} + yF_{yi} + zF_{zi} = 2F_i \quad (i = x, y, z).$$

Consequently $F_{xz}(P) = F_{zz}(P) = 0$ and $F_{yz}(P) = 2a$. Thus

$$H_F(P) = \det \begin{pmatrix} -a\phi''(0) & F_{xy}(P) & 0 \\ F_{xy}(P) & F_{yy}(P) & 2a \\ 0 & 2a & 0 \end{pmatrix} = 4a^3\phi''(0).$$

The local factorization $F(x, y, 1) = (y - \phi(x))U(x, y)$ has a holomorphic factor U with $U(0, 0) = a \neq 0$. Hence $F(x, 0, 1) = -\phi(x)U(x, 0)$, and the intersection multiplicity with $y = 0$ is the order of the zero of ϕ at 0. Because $\phi(0) = \phi'(0) = 0$, that order is at least three exactly when $\phi''(0) = 0$. It cannot exceed three: the nonzero restriction $F(x, 0, 1)$ has degree at most three. The restriction is nonzero because $y = 0$ is not a component of the irreducible cubic. Since $a \neq 0$, this proves both directions. $\square$

2 A complete calculation for the Hesse family

Consider

$$F_\lambda(x, y, z) = x^3 + y^3 + z^3 + \lambda xyz, \quad \lambda \in \mathbb{C}.$$

The sign convention here is $+\lambda xyz$; changing to the alternative convention $-3kxyz$ changes the smoothness condition accordingly.

Proposition 2.1 (Smoothness). *The curve $C_\lambda = \{F_\lambda = 0\}$ is smooth if and only if $\lambda^3 \neq -27$.*

Proof. A singular point would satisfy

$$3x^2 = -\lambda yz, \quad 3y^2 = -\lambda xz, \quad 3z^2 = -\lambda xy.$$

If one coordinate were zero, these equations would force all three to vanish, which is impossible in projective space. Multiplication and cancellation therefore give $27 = -\lambda^3$. Conversely, if $\lambda = -3\kappa$ with $\kappa^3 = 1$, the point $[1 : 1 : \kappa^{-1}]$ has all three partial derivatives zero. Euler's identity then gives $F_\lambda = 0$ there, so this is a singular point. $\square$

A smooth plane cubic over $\mathbb{C}$ is irreducible: a reducible cubic has a line factor L and a quadratic factor Q ; their restrictions have a common projective zero, and at such a point $\nabla(LQ) = 0$. If Q vanishes identically on L , the repeated line likewise gives singular points. Hence the preceding Hessian criterion applies whenever $\lambda^3 \neq -27$.

Proposition 2.2 (Nine explicit flexes). *Let $\omega = e^{2\pi i/3}$ and assume $\lambda^3 \neq -27$. The flexes of C_λ are exactly the following nine distinct points:*

$$[1 : -\omega^j : 0], \quad [1 : 0 : -\omega^j], \quad [0 : 1 : -\omega^j], \quad j = 0, 1, 2.$$

Proof. Direct differentiation gives

$$\begin{aligned} H_{F_\lambda} &= \det \begin{pmatrix} 6x & \lambda z & \lambda y \\ \lambda z & 6y & \lambda x \\ \lambda y & \lambda x & 6z \end{pmatrix} \\ &= (216 + 2\lambda^3)xyz - 6\lambda^2(x^3 + y^3 + z^3). \end{aligned}$$

In particular, the polynomial identity

$$H_{F_\lambda} + 6\lambda^2 F_\lambda = 8(27 + \lambda^3)xyz$$

shows that $F_\lambda = H_{F_\lambda} = 0$ is equivalent to

$$xyz = 0, \quad x^3 + y^3 + z^3 = 0.$$

When $z = 0$, neither x nor y can vanish. Normalize $x = 1$ to obtain $y^3 = -1$, or $y = -\omega^j$. The other two coordinate lines give the other six points. They are all distinct, since each has exactly one zero coordinate and the three cube roots are distinct. The Hessian criterion proves both that every listed point is a flex and that there are no others. $\square$

2.1 Checking one flex directly, including the Fermat cubic

At $P = [1 : -1 : 0]$, the gradient is $(3, 3, -\lambda)$, so the tangent is $3x + 3y - \lambda z = 0$. Substitute $y = -x + \lambda z/3$ without dividing by λ :

$$F_\lambda\left(x, -x + \frac{\lambda z}{3}, z\right) = \left(1 + \frac{\lambda^3}{27}\right) z^3.$$

The coefficient is nonzero under the smoothness assumption. The tangent therefore meets the cubic only at $z = 0$, namely P , with multiplicity three. This calculation remains valid for the Fermat cubic $\lambda = 0$.

3 Verification and scope

The accompanying `verify_cubic.py` checks the Hessian identity and tangent substitution symbolically in SymPy. It also verifies both equations at all nine points using the exact relation $\omega^2 + \omega + 1 = 0$, and checks the constructed singular point when $\lambda = -3\kappa$. These computations provide an independent algebra check; the proofs above establish the claims for all allowed parameters.

This note proves the local Hessian criterion and the nine-flex result for the displayed Hesse family. It does not reproduce the original report's incomplete argument for the general theorem that every smooth complex plane cubic has nine flexes. No classification of all smooth cubics into Hesse normal form is asserted here.

Source and revision record

Based on Chenrui Xu's four-page Edinburgh coursework report, *CC-R-3: Inflection Points of Smooth Cubic Curves*, archived as ED07. Its exact submission date is unconfirmed (archive period: 2021–2022). The original is retained privately. This is an expository revision dated 25 September 2026, prepared with AI-assisted mathematical and editorial checking; it is not a claim of new theorems or peer-reviewed publication.

The revision corrects the off-curve point $[1 : 1 : 0]$ to $[1 : -1 : 0]$, the signs of all nine flex coordinates, the exceptional parameter condition, and powers of λ in the example. It supplies the missing factor 2 in the mixed derivative and the missing power of $F_y(P)$ in the Hessian calculation, works with a holomorphic local chart rather than a real interval, proves both directions of the criterion, and treats $\lambda = 0$ without division by zero. The symbolic checks and explicit scope statement are new to this edition.